CVPDL HW1 Report

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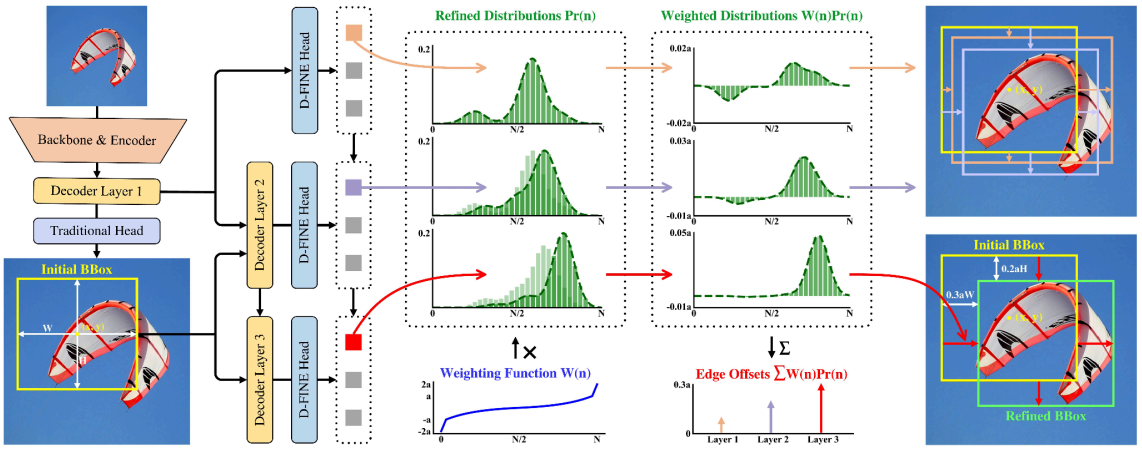
Method (Q1) :

Overview:

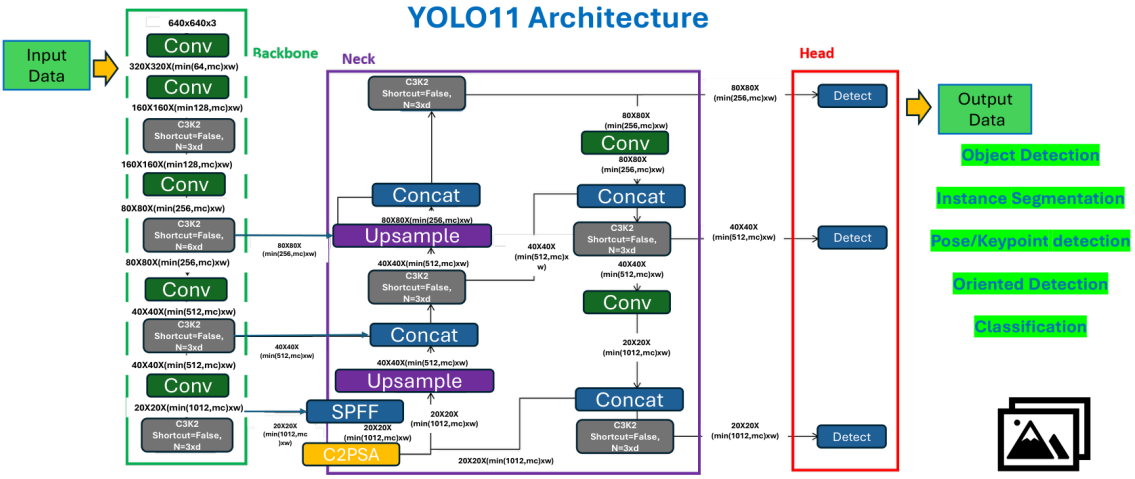
The final submission is an ensemble of six detectors. Four of them are D-FINE-L models and two are YOLO11 models (l and m). All of them start from COCO pretrained weights and are fine-tuned only on the training set. The predictions of each model are merged by class-wise NMS, and then the six models are fused by Weighted Boxes Fusion (WBF).

Model Structure:

- **D-FINE-L**



- **YOLO11-l**
- **YOLO11-m**



Loss Function:

Model	Loss
D-FINE	$L = 1 \cdot L_{VFL} + 5 \cdot L_1 + 2 \cdot L_{GIoU} + 0.15 \cdot L_{FGL} + 1.5 \cdot L_{DFL}$
YOLO11	$L = 7.5 \cdot L_{CIoU} + 0.5 \cdot L_{cls\ (BCE)} + 1.5 \cdot L_{DFL}$

- L_{VFL} : Varifocal loss for classification (α = 0.75, γ = 2), which uses the IoU as the target score.
- L_1 and L_{GIoU} : box regression losses.
- L_{FGL} : Fine-grained Localization loss on the predicted edge distributions.
- L_{DDF} : Decoupled Distillation Focal loss, used for the self-distillation (GO-LSD).
- The D-FINE loss is applied to every decoder layer, to the encoder proposals and to the denoising queries. The Hungarian matching cost is 2·class + 5·L1 + 2·GIoU.
- L_{DFL} (YOLO11) : Distribution Focal Loss on the box edges, similar idea to FDR but without refinement.

Training Settings:

	D-FINE-L	D-FINE-L	YOLO11l	YOLO11m
Input size	1536	1280	1280	1536
Epochs	72/32/72 + 4 on tiles	72	100	100
Batch	1	4	2	2
Optimizer	AdamW, lr 5e-5 (backbone 2.5e-5), wd 1.25e-4, EMA 0.9999, grad clip 0.1	AdamW, lr 1e-4 (backbone 5e-5), same others	AdamW lr 7.1e-4, cosine decay, 3 warm-up epochs, wd 5e-4	
Augmentation	photometric distortion, random IoU crop, h-flip, multi-scale (off in the last 8 epochs; last 1 epoch for the tile fine-tune)		mosaic, mixup 0.15, scale 0.7, HSV-V 0.5, h-flip	

The three D-FINE-L 1536 models are: (1) the normal 72-epoch run, (2) the same setting with only 32 epochs, and (3) model (1) fine-tuned for 4 more epochs on 640 px tiles of the training images (27,245 tiles) mixed with the 4,500 full images.

Pretrained Weights:

Weights	Source
`dfine_l_coco.pth`	github.com/Peterande/D-FINE releases
`yolo11l.pt`, `yolo11m.pt`	github.com/ultralytics/assets releases

Strategy (Q2) :

Main Path:

Main path to the final model. Each improvement is built on the row above it.

Stage	Change	val	Δ	Kaggle public
Input Scale	YOLO11s, 640 px (baseline)	0.1642	–	–
	640 → 1024 px	0.2509	+0.0867	–
	1024 → 1280 px	0.2832	+0.0323	0.25918
Model Capacity	YOLO11s → YOLO11m	0.3243	+0.0411	0.29459
Architecture	YOLO11m → D-FINE-M	0.3501	+0.0258	–
	D-FINE-M → D-FINE-L	0.3667	+0.0166	–
Inference	+ sliced 800 px + full image	0.3897	+0.0230	0.35387
Ensemble	+ 3 models, WBF	0.4213	+0.0316	0.37640
Input Scale	D-FINE-M → D-FINE-L 1536 in the ensemble	0.4262	+0.0049	0.37997
Inference	tile size and full-image scale tuned for each model	0.4306	+0.0044	0.38324
Ensemble	+ D-FINE-L 1536 with a 32-epoch schedule (5 models)	0.4331	+0.0025	0.38498

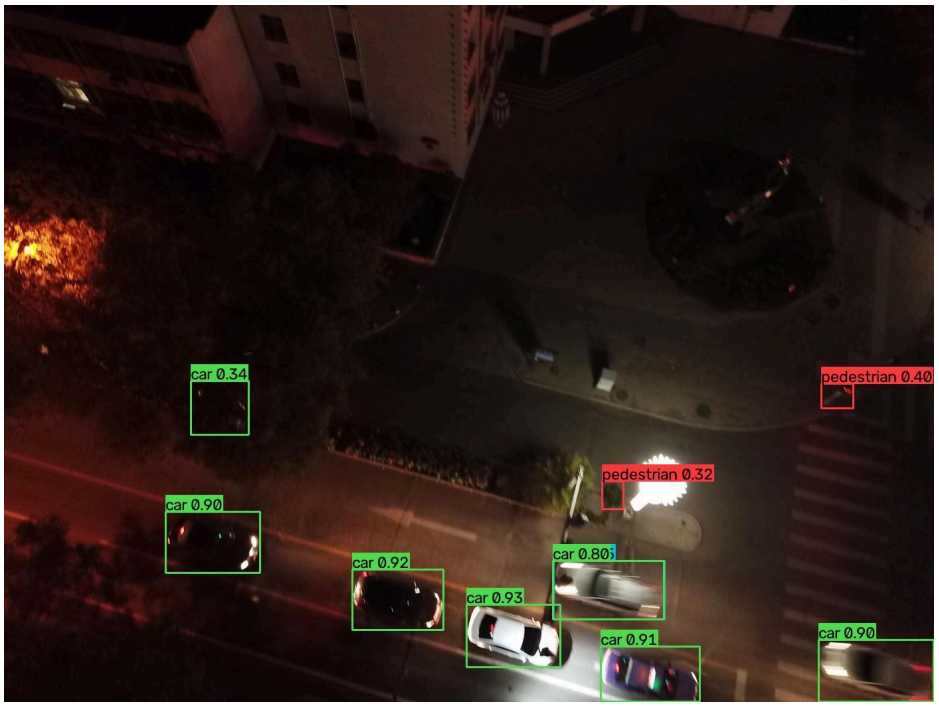
Training	+ D-FINE-L 1536 with slicing-aided fine-tuning (6 models, final)	**0.4354**	+0.0023	**0.38642**

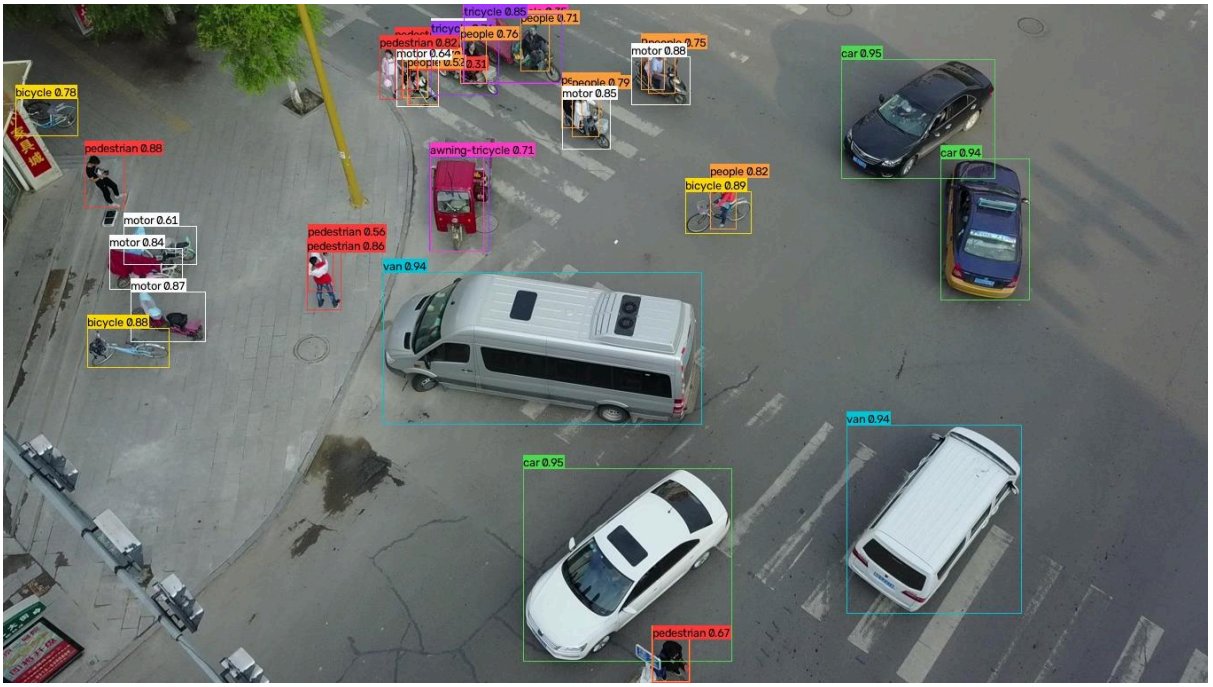
Inference Time:

Inference time strategy on 3 different architectures.

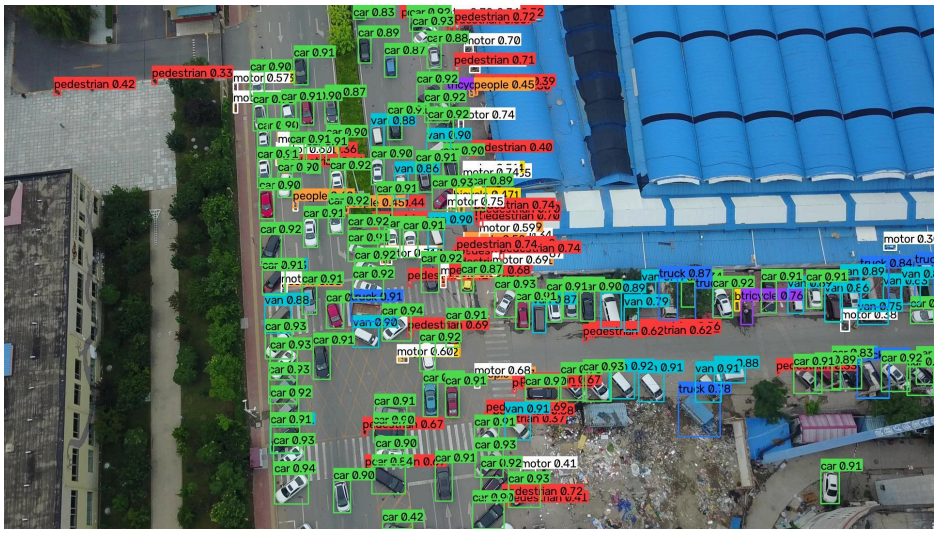
Strategy	**YOLO11m-1280**	**YOLO11l-1280**	**D-FINE-L-1280**
Default	0.3243	0.3377	0.3667
H-flip TTA (NMS)	0.3355	0.3491	0.3702
Multi-scale + h-flip	0.3545	0.3677	0.3833
Sliced 800 px + full image	0.3551	0.3728	**0.3897**
Sliced + h-flip	**0.3581**	**0.3745**	0.3890
Changing NMS IoU 0.6 → 0.5 / 0.7	±0.001	±0.001	0.0000

Visualization (Q3) :




bicycle 0.78
pedestrian 0.88
motor 0.61
motor 0.84
motor 0.87
bicycle 0.88
pedestrian 0.56
pedestrian 0.86
van 0.94
awning-tricycle 0.71
tricycle 0.85
people 0.71
people 0.76
pedestrian 0.82
motor 0.64
motor 0.85
motor 0.88
people 0.75
people 0.82
bicycle 0.89
car 0.95
car 0.94
van 0.94
car 0.95
pedestrian 0.67


car 0.35
pedestrian 0.46
pedestrian 0.41
car 0.75
car 0.60
pedestrian 0.563
pedestrian 0.47
pedestrian 0.63
people 0.60
motor 0.48
motor 0.41
people 0.70
car 0.70
pedestrian 0.40
people 0.55
people 0.67
car 0.73
car 0.87
car 0.36


pedestrian 0.42
pedestrian 0.33
motor 0.57
pedestrian 0.72
pedestrian 0.71
motor 0.70
people 0.45
motor 0.74
pedestrian 0.40
motor 0.75
pedestrian 0.74
motor 0.59
motor 0.69
pedestrian 0.74
truck 0.87
pedestrian 0.69
pedestrian 0.62
motor 0.60
motor 0.68
truck 0.78
motor 0.41
car 0.42